Agam Bansal

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Summary

Final-year B.Tech EE student with full-stack development experience (React, Node.js/Express, Flask) and a strong OOP and DSA foundation (200+ problems solved). Worked as a Backend AI Intern at FlyRank building REST APIs and RAG pipelines; prior experience shipping Flask APIs in a collaborative, Git-based workflow. Comfortable working independently in remote, cross-functional teams.

Skills

Languages: C++, JavaScript, Python, SQL
Frontend: React.js, Redux Toolkit, HTML, CSS, Tailwind CSS
Backend: Node.js, Express.js, Flask, REST APIs, JWT, Socket.IO
Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis
ORMs: Prisma, Mongoose
Tools: Docker, Git, GitHub, Postman, Cloudinary
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

Experience

Backend AI Intern July 2026 – Aug 2026
FlyRank *Remote*

- Developed backend software for AI-powered applications, including REST APIs, authentication, and data models for scalable backend services
- Contributed to Retrieval-Augmented Generation (RAG) pipelines and AI agent workflows
- Completed Anthropic Academy AI Fluency coursework to deepen practical LLM and agentic-workflow skills applied on the job

Software Developer Intern June 2025 – July 2025
Codec Technologies India *Remote*

- Developed RESTful APIs using Flask with modular architecture, role-based access control, and Git-based collaborative workflows
- Built a Python-based cryptocurrency dashboard with near real-time data updates using modular data-fetching logic

Projects

Usage Metering & Billing Engine | *Node.js, Express.js, SQLite, Stripe* GitHub

- Built a production-grade usage-metering and billing backend (FlyRank capstone), enforcing idempotent request metering, per-tenant plan quotas, and Stripe-synced subscription billing
- Designed integer nano-dollar cost math for AI-token pricing (cached-input as a subset, reasoning billed at output rate), eliminating floating-point drift across usage rollups
- Implemented signature-verified, replay-proof Stripe webhook handling with idempotency-key deduplication and a UNIQUE-constraint safety net against concurrent-retry races
- Designed a layered architecture (routes → services → db) so core metering, quota, and cost logic run with zero external dependencies, fully offline and swappable to a different database engine
- Validated with 35/35 automated tests covering idempotency, quota boundaries, pinned cost math, and webhook verify/dedup

Production-Ready URL Shortener | *Node.js, TypeScript, Express.js, PostgreSQL, Prisma, Redis* GitHub

- Built a production-grade URL shortener with custom aliases, URL expiration, and REST APIs backed by a Redis cache-aside architecture
- Implemented cache-aside caching with write invalidation, cutting average redirect latency from ~59ms to ~5ms (~11x speedup) and reducing database queries by ~99%
- Built an atomic, race-condition-free rate limiter using a Redis sorted-set sliding-window algorithm via Lua scripting, validated at 62/150 concurrent requests correctly blocked at 100 req/min
- Ensured race-condition-safe short code generation using a database unique constraint with a retry loop on collision (P2002), guaranteeing no duplicate aliases under concurrent writes
- Increased redirect throughput from ~5K to ~50K requests/sec through in-memory caching, TTL-based cache invalidation, and efficient database access

Education

Dr B.R. Ambedkar National Institute of Technology, Jalandhar Jalandhar, India
B.Tech in Electrical Engineering, CGPA: 7.89 *Expected 2027*

Problem Solving

Solved 200+ DSA problems on LeetCode with strong proficiency in Dynamic Programming, Graphs, Trees, and Backtracking.